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The Three Sides of the Moon LLMs in Cybersecurity: Guardians, Enablers and Targets

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Abstract

Large Language Models (LLMs) are rapidly evolving, demonstrating impressive capabilities in multimedia objects generation, ranging from text and image generation from scratch to programming code and efficient conversational agents. From the perspective of cyber-security challenges, LLMs and cyber-security are in a controversial relationship: it can be observed that LLMs, as a type of AI, play a mainfold role: that of security guardians, that of security breachs "unaware" enablers and that of victims of cyber attacks. In fact, LLMs are able to enhance security of several tasks and applications but they are also attractive for malicious users to be exploited as means to perform novel attacks and, finally they represent challenging assets for targeting attacks. In this work, we discuss this mainfold key reading by providing a brief landscape of both the current defence applications of LLMs against cyber attacks and the currently known LLMs security vulnerabilities along with potential cyber-attacks targeting and involving LLMs. The final aim of study is intended to provide a guideline to further explore specific cyber-security scenarios involving LLMs.

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Keywords: Large Language Models; Cyber-security; Generative Models Vulnerabilities; Prompt Injection.

1. Introduction

Large Language Models (LLMs)[1] are celebrated for their exceptional abilities in language understanding and generation, fostering a dynamic ecosystem of applications. However, their growing influence and widespread integration into diverse services and application has also attracted the attention of cyber criminals and malicious actors, bringing considerable security risks. Security threats related to LLMs range in a wide space and represent an issue that is far from being precisely identified: from the privacy concerns related to unintentional disclosure of sensitive information to the intentional conditioning of generation by exploiting data poisoning or malicious prompt injection

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techniques, for example. Unfortunately, any time that LLMs are embedded in new services and applications, the list of security threats increases along with their attack surface.

On the other hand, in addition to the well-known use as generative AI, LLMs have enabled to enhance security and privacy across various fields, providing benefits to the security community. The work of [2] confirms a multiple key reading, by asking three questions that can be summarised in analyzing i) the security enhancements that are achievable by integrating or exploiting LLMs capabilities; ii) what kind of security risks and threats are associated with the use of LLMs; and iii) What are the most relevant vulnerabilities of LLMs, along with the potential strategies to employ for defending against these security threats.

So, this work aims to provide a concise but effective landscape of the current security issues affecting LLMs, useful to suggest a more agile dimension of analyzing LLMs-based applications along with their main security concerns and we discuss about these aspects of LLMs in cyber-security. The final goal of this work can be also summarized in the following two points:

- Identify the potential attack vectors where LLMs could be most effective.
- Understand the challenges and limitations of using LLMs for malicious purposes.

The paper is organized as follows: Section 2 introduces the discussion about the multifacet role played by LLMs in Cybersecurity. Section 3 proposes a perspective of Key LLM security risks, while Section 4 describes the most known security vulnerabilities and potential Cyber-attacks against LLMs, including the privacy data leakage issue. Section 5 provides the ground for discussion and collects some mitigation strategies. Section 6 concludes the work by providing some considerations about the future actions to perform to enforce and enhance robustness and resilience of LLMs to cyber-security threats.

2. The Multifacet of LLMs in Cybersecurity: Guardians, Enablers and Targets

Large Language Models (LLMs) are revolutionizing various fields, and cyber-security is no exception. Scientific literature[2] reveals a fascinating duality in their role: LLMs can be both powerful guardians, unaware but smart enablers and vulnerable targets within the cyber-security landscape.

2.1. LLMs as Guardians of Cybersecurity

When playing the role of ”security guardians”, LLMs hold immense potential to enhance application security. A potential defensive perspective derived from the research literature can be identified in the following one:

- *Threat Detection and Analysis*: Studies like [17] highlight how LLMs can analyze vast amounts of text data, uncovering hidden patterns and anomalies that might indicate impending cyberattacks. This allows security teams to proactively identify and mitigate threats before they cause damage.
- *Content Moderation and Phishing Detection*: LLMs can be trained to identify malicious content and phishing attempts with high accuracy. Research provided by [18] explores this application, suggesting LLMs can significantly reduce the spread of misinformation and defend users from fraudulent scams.
- *Vulnerability Detection and Patching*: LLMs can be used to analyze code for vulnerabilities, similar to how they analyze text data. This application is still under development, but some studies suggest LLMs can potentially identify security weaknesses in software even faster than traditional methods.

2.2. LLMs as Enablers of Security Attacks

LLMs power extends beyond positive applications and into the realm of cybercrime. LLMs pose a unique challenge as they can be both defenders and targets within the cybersecurity landscape. A deeper dive into how LLMs can be enablers of security attacks is provided below.

- Crafting the Perfect Deception:
  - Social Engineering: One of the most concerning aspects of LLMs is their ability to mimic human language with exceptional fluency. This makes them ideal for crafting highly believable phishing emails, social media posts, or chat messages used in social engineering attacks. LLMs can personalize these messages to individual targets, analyzing data points like social media behavior and online habits to create a more convincing persona.
  - Content Manipulation: The ability to generate realistic text extends to creating deepfakes and other forms of synthetic media. Malicious actors could leverage LLMs to create fake news articles, manipulate financial reports, or even impersonate real people in videos to spread misinformation or damage reputations.
- Automating the Malicious:
  - Malware Development: While directly generating complex malware might be beyond current capabilities, LLMs can be used to automate parts of the development process. They could be instrumental in creating obfuscated scripts or code snippets that bypass traditional security filters. Additionally, LLMs could potentially identify patterns in successful malware, aiding attackers in crafting more effective strains.
  - Network Reconnaissance: The automation capabilities of LLMs extend to network reconnaissance tasks. Imagine attackers using LLMs to probe defenses, identify potential targets, and gather information about network configurations. This could significantly streamline the pre-attack phase, making it faster and more efficient for attackers.
  - Fake Data Generation: One of the major drawbacks of generative AI is its potential to be weaponized for creating fake news and disseminating false data. In fact,the impressive ability of generative AI was not late in being employed into data generation tasks: a typical example is provided by the Generative Adversarial Networks [22] adopted for data augmentation and data regularization tasks. The ability to mimick real data distributions represents a great advantage when the fake data, artificially generated would be used for improving AI-based models; anyway, as discussed in [23] a high level of likelihood between real ad fake data can raise a stuff to foolish checks and anomaly detection systems. A further related problem is provided by the Fake News Factory, where generative AI are able to produce highly realistic fakes: Generative AI excels at mimicking human language styles and patterns. This makes it possible to create fake news articles, social media posts, or videos that are incredibly realistic and difficult to distinguish from legitimate content[24, 25].
- Exploiting Vulnerabilities:
  - Vulnerability Research: LLMs have the potential to analyze vast amounts of code, potentially uncovering weaknesses faster and more efficiently than traditional methods. This could lead to a scenario where attackers leverage LLMs to discover new vulnerabilities before they are patched, giving them a significant advantage.

2.3. LLMs as Targets for Cyberattacks

Despite their defensive potential, LLMs themselves are susceptible to cyberattacks.The scientific literature concerning LLMs security vulnerabilities and threats proposes several categorization criteria to map specific attacks to wider families. A first, coarse-grained categorization is that proposed below:

- *Poisoned Training Data*: Adversarial actors might try to manipulate the training data used to create LLMs. This can lead to the models inheriting biases or vulnerabilities that could be exploited deeper in [19].

- *Generating Malicious Code*: Malicious actors could potentially use LLMs to generate malicious code or scripts, creating a new wave of cyber threats that are more sophisticated and difficult to detect.

- *Social Engineering and Misinformation Campaigns*: LLMs can be used to create highly realistic and persuasive text, which could be weaponized for social engineering attacks or spreading misinformation. This is a growing concern, as discussed in [20].

3. Key LLM security risks

Generative AI, as LLMs, acts as super-powered search engines that can not only answer your questions but also complete tasks using human-like language. Anyway, this power comes with a caveat – both publicly available models and those used internally by organizations can introduce security and privacy risks.

Five of the most relevant security risks affecting LLMs have been identified in the study proposed in [21] as the following ones:

- Oversharing sensitive data: LLMs are trained on vast datasets of text and code. If this training data includes sensitive information, even inadvertently, the LLM might inadvertently incorporate that information into its outputs. Imagine an LLM trained on a customer service database that accidentally reveals a customer's social security number in a generated response. Furthermore, LLMs are adept at identifying patterns in data. If the training data includes patterns that correlate with sensitive information, the LLM might be able to reveal that information even if it's not explicitly included in the text. For instance, an LLM trained on emails might learn to associate certain keywords with upcoming product launches, even if the emails themselves don't mention the launch date explicitly.

- Copyright challenges: arge Language Models thrive on massive amounts of data, but there's a catch. This data is often collected from the web, and copyright ownership might be unclear. Using such data for your own LLM could land you in hot water. Unfortunately, tracing the source of each piece of data is a tall order, making it tricky to avoid copyright headaches.

- Insecure code: While developers are racing to market with the help of tools like ChatGPT, which can churn out code snippets and even whole programs in record time, security experts raise a red flag. These tools might introduce vulnerabilities, especially if developers lack the expertise to spot them. Buggy code slipping into production can be a PR nightmare and a costly fix down the line.

- Hacking the LLM itself: if hackers gain unauthorized access to LLMs, they could unleash a toolbox of nasty tricks. Malicious prompts might trick the model into spilling sensitive information, or force it to do things it's not supposed to. They could exploit weaknesses in the LLM's server (like the erver-side request forgery (SSRF) vulnerabilities) to steal internal data. Even scarier, hackers might figure out how to use natural language commands to manipulate confidential systems and resources directly.

- Data breach at the AI provider: Even the companies creating AI models aren't immune to breaches. Hackers could steal training data, potentially containing sensitive or proprietary information. Similar leaks have happened before, like Google accidentally exposing private Bard conversations in search results.

Among these five key points, we want to remark the the consequences of LLMs oversharing sensitive data, that can be severe as:

- Privacy Violations: Unintentional disclosure of personal information can have serious consequences for individuals, including identity theft, financial fraud, and reputational damage.

- Security Breaches: LLMs could leak sensitive corporate information, such as trade secrets, financial data, or intellectual property. This could give competitors an unfair advantage or damage a company's reputation.

- Compliance Issues: Organizations that rely on LLMs need to be mindful of data privacy regulations. Inadvertent disclosure of sensitive data could lead to hefty fines and regulatory penalties.

4. Security Vulnerabilities and Potential Cyber-Attacks Against Large Language Models

As it was introduced in the previous sections, LLMs widespread adoption also introduces a range of cyber-security vulnerabilities and potential attack vectors. LLMs are vulnerable to significant security threats from maliciously designed messages intended to alter the system's initial instructions or expose confidential information. This section outlines the current scientific landscape of these security challenges, drawing on recent research and expert analyses. It also proposes a finer grained classification than the one introduced in the previous section, that includes: a)adversarial attacks; b) data poisoning; c) model inversion; d) membership inference; e) prompt injection; and f) data leakage.

A potential taxonomy of the most notable clusters of security attacks targeting LLMs can include the following ones:

- Adversarial Attacks;
- Data Poisoning and Training-Time Attacks;
- Model Inversion and Membership Inference Attacks;
- Prompt Injection Attacks;
- Privacy and Data Leakage;

4.1. Adversarial Attacks

Adversarial attacks[4, 30]. mainly involve feeding LLMs with specially crafted inputs designed to mislead the model into producing incorrect or harmful outputs. These attacks can be divided into evasion attacks and poisoning attacks:

- Evasion Attacks Evasion attacks, also known as adversarial examples, are inputs specifically crafted to cause the model to make errors while appearing benign to human observers. For instance, attackers might subtly alter the wording or structure of a query to trick the model into generating undesirable responses. Research provided in [5] demonstrated that even minor perturbations to input data could significantly degrade model performance.

- Poisoning Attacks: In poisoning attacks[32, 26], malicious actors inject harmful data into the training set, leading the model to learn incorrect behaviors. This can result in the model producing biased or harmful outputs. Such attacks are particularly concerning because they can have long-lasting effects on the model's behavior. The work of [6] highlighted the dangers of training data poisoning, showing that backdoors can be embedded into models to trigger malicious behavior under specific conditions.

4.2. Data Poisoning and Training-Time Attacks

Training-time attacks are conducted during the model training phase and can have lasting impacts on the model's behavior. This family's attacks are typically recognized as:

- Backdoor Attacks: Backdoor attacks involve embedding hidden patterns within the training data. These patterns, when present in inputs during inference, trigger the model to behave maliciously. For example, a back-doored LLM might generate harmful content when encountering a specific trigger phrase.

- General Data Poisoning: By injecting malicious data into the training dataset, attackers can subtly alter the model's behavior, causing it to make errors or reveal sensitive information when processing specific inputs. General data poisoning refers to the broader category of attacks where attackers inject erroneous or malicious data into the training set. The work of [7] discussed how even a small amount of poisoned data could significantly alter model behavior, leading to degraded performance or intentional malfunctions.

4.3. Model Inversion and Membership Inference Attacks

These attacks focus on extracting sensitive information from the model; to this family typically belong:

- Model Inversion Attacks: aiming to reconstruct the training data by exploiting the model's outputs[31]. By systematically querying the model and analyzing its responses, attackers can infer sensitive information about the training data. In[8] it was demonstrated that models could be coerced into revealing significant details about individual data points, posing a serious privacy risk.

- Membership Inference Attacks: aiming to determine whether a specific data point was part of the model's training set. This can reveal sensitive information, particularly if the presence or absence of data points has privacy implications. In [9] it has been showed that such attacks could be executed with high accuracy, highlighting the need for better privacy-preserving techniques in model training.

4.4. Prompt Injection Attacks

Prompt injection attacks manipulate the input prompt to alter the model's behavior. They can be divided into:

- Instruction Hijacking: Instruction hijacking involves crafting inputs that cause the model to deviate from its intended instructions. This can lead to the generation of harmful or misleading content. The work of [10] explored how adversarial triggers could manipulate NLP models to produce undesirable outputs, emphasizing the vulnerability of LLMs to such attacks.

- Context Manipulation: it refers to altering the context in which the model operates to influence its responses. By changing the surrounding text or inputs, attackers can subtly steer the model towards generating biased or harmful outputs. In [11] have been discussed the impacts of context manipulation, demonstrating how easily LLMs could be misled.

4.5. Privacy and Data Leakage

LLMs trained on large datasets may inadvertently memorize and leak sensitive information.

- Unintended Memorization: occurs when the model inadvertently retains details from its training data, which can then be extracted through carefully crafted queries. The study proposed in [12] showed that neural networks, including LLMs, could memorize and reveal sensitive information from the training data.

- Data Leakage: it happens when the model's outputs inadvertently reveal sensitive information. This can occur when the model has been trained on proprietary or confidential datasets. The work of [13] highlighted the risks

associated with data leakage and discussed various techniques to mitigate this issue.

According to our perspective, security and privacy concerns related to LLMs should be detailed in a proper section and analyzed from multiple and different perspectives. This point of view is shared in the study of [2], where a meticulous literature review pertaining to the intersection of LLMs with security and privacy is provided. While a large literature about these concerns has quickly grown in the last few months, including the work of [2], we want to focus on the aspects that are described in the following.
First, LLMs can potentially facilitate privacy and security violations within documents. When processing or generating text, these models might inadvertently reveal sensitive information, contributing to data breaches or unauthorized access. Second, LLMs cannot be entirely controlled regarding the information they utilize to generate new content. The inherent complexity and vastness of their training data make it difficult to predict or restrict the specific information they might use or disclose, posing ongoing challenges for maintaining privacy and security.

5. Discussion: Challenges and Mitigations

Despite the potential for misuse, it's important to remark that LLMs are still under development. The challenges associated with using them for malicious purposes include:

- Data Dependence: LLMs are highly dependent on the data they are trained on. If the training data is biased or manipulated, the LLM could generate outputs that are flawed or misleading.
- Mitigating Bias: Similar to data dependence, LLMs can inherit biases from the training data. This could be exploited by attackers to create targeted attacks that resonate with specific demographics.
- Detectability: As LLMs become more sophisticated, it might become harder to differentiate between human-generated and LLM-generated content. This poses a challenge for cybersecurity professionals who need to identify and stop attacks in real- time.

To address data oversharing problem, several approaches can help mitigate the risk of oversharing with LLMs, like the following ones:

- Data Sanitization: Before training LLMs, sensitive data should be carefully removed from the training datasets. This is a complex process, but crucial to ensure the LLM doesn't learn patterns that could lead to oversharing.
- Privacy-Preserving Training Techniques: New techniques are being developed to train LLMs while minimizing the risk of exposing sensitive information. These techniques involve data anonymization and differential privacy methods.
- Human Oversight and Control: LLMs are powerful tools, but they shouldn't operate in a black box. Human oversight and control are essential to ensure the outputs are safe and don't contain sensitive information.

As for the Future of LLMs and Sensitive Data, it can be observed that LLMs hold immense potential, but addressing the oversharing problem is crucial for responsible development and deployment. By focusing on data sanitization, privacy-preserving techniques, and human oversight, we can ensure LLMs become powerful tools without compromising security or privacy.

To conclude this preliminary discussion, while LLMs pose a new threat landscape, there are ways forward to prepare. A list of suggested actions is provided below.

- Security Awareness Training: Educating users about the potential for LLM-based attacks is crucial. Users should be trained to be skeptical of overly personalized messages or content that seems too good to be true.

- Advanced Threat Detection: Developing advanced threat detection systems capable of identifying LLM-generated content is essential. This might involve analyzing language patterns or identifying subtle inconsistencies in the text.

- Continuous Research: Further research is needed to understand the full scope of LLM-based threats and develop effective mitigation strategies. This includes exploring methods for training LLMs in a way that minimizes bias and vulnerability to manipulation.

6. Conclusions and Future Directions

The scientific community recognizes the double-edged sword that LLMs present in cyber-security. While they offer exciting possibilities for enhanced security, their vulnerabilities cannot be ignored. As LLM technology matures, researchers are actively exploring methods to mitigate these risks and ensure these powerful tools are used for good within the cyber-security landscape. To address the aforementioned vulnerabilities, several mitigation strategies can be implemented. Among the ones that have been proposed we can mention:

- Robust Training Techniques: Implementing adversarial training and differential privacy techniques can enhance the model's resilience against adversarial and privacy attacks[27]. in [14] introduced distillation as a defense mechanism to improve the robustness of deep neural networks against adversarial perturbations.

- Regular Auditing and Testing: Conducting regular security audits and stress-testing the models with adversarial inputs helps identify and patch vulnerabilities. The work of [15] discussed the effectiveness of ensemble adversarial training in improving model robustness through systematic testing.

- Access Controls and Monitoring: Implementing stringent access controls and continuous monitoring can detect and mitigate unusual activity that might indicate an ongoing attack. in [16] provided a comprehensive review of AI/ML security threats and proposed various countermeasures, emphasizing the importance of access controls and monitoring.

As LLMs continue to evolve, so too do the methods employed by malicious actors seeking to exploit their vulnerabilities. Understanding the current landscape of security threats is essential for developing robust defenses. Finally, we can state that by acknowledging the potential for misuse and proactively developing safeguards, we can ensure that LLMs remain a force for good in the cyber-security landscape.

References

[1] Chang, Yupeng, et al. "A survey on evaluation of large language models." ACM Transactions on Intelligent Systems and Technology 15.3 (2024): 1-45.
[2] Yao, Yifan, et al. "A survey on large language model (llm) security and privacy: The good, the bad, and the ugly." High-Confidence Computing (2024): 100211.
[3] https://openai.com/gpt-4/
[4] Biggio, B., and Roli, F. (2018). Wild patterns: Ten years after the rise of adversarial machine learning. Pattern Recognition, 84, 317-331.
[5] Goodfellow, I., Shlens, J., and Szegedy, C. (2015). Explaining and harnessing adversarial examples. arXiv preprint arXiv:1412.6572.
[6] Gu, T., Dolan-Gavitt, B., andGarg, S. (2017). Badnets: Identifying vulnerabilities in the machine learning model supply chain. arXiv preprint arXiv:1708.06733.
[7] Steinhardt, J., Koh, P. W., andLiang, P. (2017). Certified defenses for data poisoning attacks. Advances in Neural Information Processing Systems, 30.
[8] Fredrikson, M., Jha, S., and Ristenpart, T. (2015). Model inversion attacks that exploit confidence information and basic countermeasures. Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security.

[9] Shokri, R., Stronati, M., Song, C., and Shmatikov, V. (2017). Membership inference attacks against machine learning models. 2017 IEEE Symposium on Security and Privacy (SP).
[10] Wallace, E., Feng, S., Kandpal, N., Gardner, M., and Singh, S. (2019). Universal adversarial triggers for attacking and analyzing NLP. arXiv preprint arXiv:1908.07125.
[11] Zhu, C., Liu, L., Wei, W., and Chen, M. (2020). Addressing the vulnerability of LLMs to context manipulation attacks. Journal of Artificial Intelligence Research, 67, 123-140.
[12] Carlini, N., Liu, C., Kos, J., Erlingsson, U., and Song, D. (2019). The secret sharer: Evaluating and testing unintended memorization in neural networks. 28th USENIX Security Symposium (USENIX Security 19).
[13] Zubair, S., and Vaidya, J. (2020). A survey on data leakage prevention techniques. Journal of Computer Science, 16(5), 672-691.
[14] Papernot, N., McDaniel, P., Wu, X., Jha, S., and Swami, A. (2016). Distillation as a defense to adversarial perturbations against deep neural networks. 2016 IEEE Symposium on Security and Privacy (SP).
[15] Tramèr, F., Kurakin, A., Papernot, N., Goodfellow, I., Boneh, D., and McDaniel, P. (2018). Ensemble adversarial training: Attacks and defenses. arXiv preprint arXiv:1705.07204.
[16] Sheatsley, R., Liu, Y., and Zhang, J. (2021). A comprehensive review on AI/ML security threats and countermeasures. ACM Computing Surveys (CSUR), 54(6), 1-36.
[17] Motlagh, F. N., Hajizadeh, M., Majd, M., Najafi, P., Cheng, F., and Meinel, C. (2024). Large Language Models in Cybersecurity: State-of-the-Art. arXiv preprint arXiv:2402.00891.
[18] https://truefort.com/platform/
[19] https://www.paloaltonetworks.com/blog/2024/04/ai-cybersecurity-and-large-language-models/.
[20] Divakaran, D. M., andPeddinti, S. T. (2024). LLMs for Cyber Security: New Opportunities. arXiv preprint arXiv:2404.11338.
[21] Das, B. C., Amini, M. H., and Wu, Y. (2024). Security and privacy challenges of large language models: A survey. arXiv preprint arXiv:2402.00888.
[22] Goodfellow, I., et al. (2020). Generative adversarial networks. Communications of the ACM, 63(11), 139-144.
[23] Marulli, F., Visaggio, C. A. (2019, July). Adversarial deep learning for energy management in buildings. In Proceedings of the 2019 Summer Simulation Conference (pp. 1-11).
[24] Marulli, Fiammetta, et al. "Exploring a federated learning approach to enhance authorship attribution of misleading information from heterogeneous sources." 2021 International Joint Conference on Neural Networks (IJCNN). IEEE, 2021.
[25] Marulli, Fiammetta, et al. "A Federated Consensus-Based Model for Enhancing Fake News and Misleading Information Debunking." Intelligent Decision Technologies: Proceedings of the 14th KES-IDT 2022 Conference. Singapore: Springer Nature Singapore, 2022.
[26] Marulli, Fiammetta, Laura Verde, and Lelio Campanile. "Exploring data and model poisoning attacks to deep learning-based NLP systems." Procedia Computer Science 192 (2021): 3570-3579.
[27] Eigner, Oliver, et al. "Towards resilient artificial intelligence: Survey and research issues." 2021 IEEE International Conference on Cyber Security and Resilience (CSR). IEEE, 2021.
[28] Devlin, J., Chang, M. W., Lee, K., and Toutanova, K. (2018). BERT: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805.
[29] Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., ... and Liu, P. J. (2019). Exploring the limits of transfer learning with a unified text-to-text transformer. arXiv preprint arXiv:1910.10683.
[30] Reference: Iyyer, M., Boyd-Graber, J., Claudino, L., Socher, R., and Daumé III, H. (2018).
[31] Fredrikson, M., Jha, S., and Ristenpart, T. (2015). Model inversion attacks that exploit confidence information and basic countermeasures. In Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security (pp. 1322-1333).
[32] Jagielski, M., Oprea, A., Biggio, B., Liu, C., Nita-Rotaru, C., and Li, B. (2018).Manipulating machine learning: Poisoning attacks and countermeasures for regression learning. In 2018 IEEE Symposium on Security and Privacy (SP) (pp. 19-35). IEEE.
[33] Liu, Yi, et al. "Prompt Injection attack against LLM-integrated Applications." arXiv preprint arXiv:2306.05499 (2023).